

Exhibit CC-45

A Novel Simulation Methodology Reflecting System Power Scenario using a Markov-Chain-based Stochastic Random Power Model

Woo-Jin Na

Memory Division
Samsung Electronics
Hwaseong-si, Korea
wj.na@samsung.com

Kun Joo

Memory Division
Samsung Electronics
Hwaseong-si, Korea
kun.joo@samsung.com

Rak-Joo Sung

Memory Division
Samsung Electronics
Hwaseong-si, Korea
rakjoo.sung@samsung.com

Kyu-Dong Lee

Memory Division
Samsung Electronics
Hwaseong-si, Korea
kona.lee@samsung.com

Ji-Hye Yang

Memory Division
Samsung Electronics
Hwaseong-si, Korea
jihye97.yang@samsung.com

Kyung-Sun Kim

Memory Division
Samsung Electronics
Hwaseong-si, Korea
sun_a@samsung.com

Young-Ho Lee

Memory Division
Samsung Electronics
Hwaseong-si, Korea
youngho05.lee@samsung.com

Seung-Hee Mun

Memory Division
Samsung Electronics
Hwaseong-si, Korea
hanmoe28@samsung.com

Sung-Joo Park

Memory Division
Samsung Electronics
Hwaseong-si, Korea
sjoo@samsung.com

Jeong-Hyeon Cho

Memory Division
Samsung Electronics
Hwaseong-si, Korea
caleb1@samsung.com

Abstract—In this paper, we present a novel power integrity simulation method based on Markov-Chain-based Stochastic Random Power Model (MCSRPM). In this method, we analyze numerous memory traffic cases of an actual system, create a Markov-Chain table which can reflect in system memory scheduling, and create MCSRPM by applying it to DDR5 DRAM state diagram. In practice, we have successfully verified that MCSRPM can predict power noise measured on a DDR5 server system. Also, it is possible to verify many power scenarios predictable in DDR5 systems at once, and confirmed that the analysis time has reduced by more than 60% compared to CPM-based simulation which can only verify few power scenarios.

Keywords—component; Power Integrity; Power Model; CPM; Current Profile; DRAM; Markov-Chain;

I. INTRODUCTION

Triggered by the increasing market demand for high performance and low-power devices with the advent of artificial intelligence (A.I) and fourth industrial revolution, the data rate as well as the capacity of Double Data Rate (DDR) memory modules keeps on increasing, while the supply voltage keeps on decreasing [1]. The data rate for DDR5 is expected to reach up to 7.2Gbps, which is twice that of DDR4 and density for DDR5 Dual Inline Memory Module (DIMM) reached up to 512GB which is twice that of DDR4. In other words, guaranteeing the same level of power integrity of DIMM is becoming extremely difficult. Moreover, some new schemes are newly adopted to DDR5

DIMM for both system stability and high performance such as Same Bank Refresh (SBR), Decision Feedback Equalizer (DFE) and on-die ECC (Error Correcting Code) which make power integrity worse. Therefore, it is critical for reliable memory operation to be guaranteed in various power scenarios in real system. Today, CPM-based simulation is widely used to simulate the power integrity of DIMM or other set of devices. However, CPM has several drawbacks for making the simulation accurate and faster. First, current profile in a CPM, which is PWL (Piece-wise Linear) model, is very fine compared to switching frequency of PMIC (Power Management Integrated Circuit). It is around several pico-seconds scale, which makes extracting and simulation time longer and user could verify only a few power scenarios. Second, current profile in a CPM is not a measured data but simulation data. It is extremely difficult to measure several pico-seconds scaled current profile because there is no equipment which is able to measure it accurately. Strictly, it is not a verified data. Third, current profile is usually extracted from very early version of device, which means it couldn't reflect the device of mass production version in simulation. In order to overcome these shortcomings of CPM, this paper proposes a new methodology using average current depending on DDR5 operation modes. As a part of continuous evolution of DDR, DDR5 has newly adopted (Switching Mode Power Supply) type PMIC on DIMM, which means DRAM uses regulated power which PMIC on DIMM supplies on DIMM. However, switching frequency of PMIC is relatively even lower than that of CPM. DDR5

PMIC switches by 0.5~1.25MHz while CPM has several GHz frequency components, which means PMIC cannot react such a fast current profile. In other words, pico-seconds scaled current changes (di/dt) couldn't affect the power noise. In order to verify that, we compared the simulation results with between current profile and DRAM average current which is called IDD and 90% consistency is verified. Based on the consistency, multiple power scenarios could be modeled since power integrity simulation with the transition of average current is even faster than that of CPM. Therefore, this paper modeled DRAM state diagram to Markov-chain, in which we applied probability to transition between DRAM states in order to reflect power scenario in an actual DDR5 system. In order to do that, we developed the software which could analyze singularity for DDR5 memory traffic and then we could get "DDR5 system Markov-chain parameter" through analyzing tons of memory access traffic cases of a real DDR5 system using it. Finally, in the power integrity point of view, we verified and predicted more than 90% consistency with voltage peak to peak noise measured in DDR5 system, when we applied DDR5 system Markov-chain parameter to MCSRPM and simulated with it. Proposed methodology proved that tons of system power scenarios could be reflected with MCSRPM and verified its practicality through making simulation time reduce by 60% while CPM-based simulation could verify only a few power scenarios.

II. CONCEPT OF DDR5 POWER INTEGRITY SIMULATION

Main purpose of guaranteeing PI for the DIMM is to optimize the design of PDN (Power Distribution Network) including decoupling capacitors and behavior of current so that stable voltage can be supplied throughout a DIMM. However, there is a key difference between DDR4 and DDR5 DIMM. In DDR4 system, PMIC is located in a system board and in charge of supplying and regulating power to six or eight DIMMs and memory interface of CPU on board. In other words, they share the same single PDN in a system board, which means that they could be affected each other by things like SSN (Simultaneous Switching Noise) and fluctuation of the power. Therefore, it is very difficult to predict the quality of PI for DDR4 DIMM in a system since it is extremely difficult to predict DRAM operation in independent channels. In the era of DDR5, the most significant change is that PMIC is adopted on DIMM called VR (Voltage Regulator) on DIMM. PMIC moved from system board to DIMM and is able to provide power to devices on DIMM such as DRAM, RCD (Register Clock Driver), SPD (Serial Presence Detect), TS (Temperature Sensor) as shown in Figure 1. DDR5 DIMM is much less affected by adjacent DIMM and CPU in terms of PI since DDR5 DIMM has its own PMIC respectively and isolated PDN. Also it makes PI simulation more accurate than that of DDR4. Figure 2 shows simulation deck structure for the DDR5 RDIMM (Registered Dual In-line Memory Module), in which there are PMIC as a voltage regulator, decoupling capacitor, DIMM PCB (Printed Circuit Board), DRAM package, RCD, RCD die and DRAM die. User simulates the

PI of DIMM and guarantees stability of each power net by applying power model to the DRAM and RCD die [2].

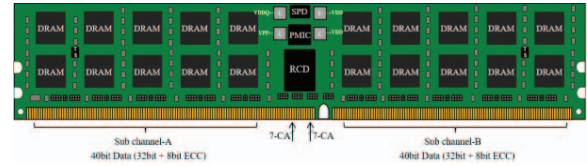


Figure 1. DDR5 RDIMM 2Rx4 with newly introduced PMIC.

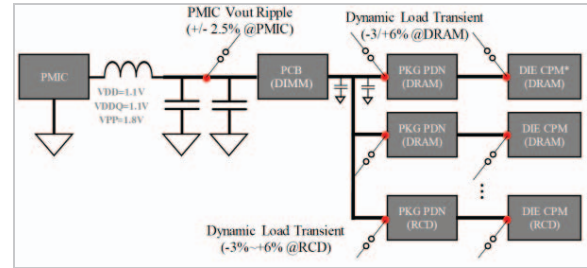


Figure 2. Simple PI simulation deck for DDR5 RDIMM [2].

III. NEW EXPERIMENT TO ENHANCE THE ACCURACY OF PI SIMULATION AND MOTIVATION

DIMM is a common type of memory module and operates by grouping DRAM components on a PCB that work together to perform commands and store data requested by the host or CPU. This is called rank, and DDR5 DIMM consists of 10 components per rank in the case of x4 RDIMM and 5 components per rank in the case of x8 RDIMM. For instance, if the host sends write command to the first rank of the DIMM, all 10 DRAM components in the first rank receive the command and data transmitted by the host and then store them to the same row/column address in the DRAMs at the first rank respectively. As a result, each DRAM component in a rank receive the same command and perform the same operation and consume the same transient power. Therefore, the PI simulation of memory module is performed by summing the current profile of DRAM as much as the number of components. However, what should be remembered is DRAM components in a rank receive command through the DIMM PCB. For improvement of SI, command and address signals in DIMM started to use fly-by topology since DDR3. DRAM components on DIMM PCB receive command sequentially through PCB routing, which means the timing of receiving command and address varies slightly depending on the location of DRAMs on PCB. It also means starting point of operating timing in DRAMs varies due to the PCB routing. In this chapter we analyzed the effect of peak current distribution due to PCB delay and verified if it need to be applied or not.

A. The effect of peak current distribution due to PCB delay

From the conventional PI point of view, current profile should be summed with timing delay applied to each DRAM components due to the signal length on PCB. Figure 3 shows

the difference in command arrival time depending on position of DRAM components in DDR5 2Rx4 DIMM. The delays shown in Figure 3 were calculated based on the read training values extracted from DDR5 ATE (Automated Test Equipment). That is, it was calculated using the difference between the time read command was applied to DRAMs from the ATE and the time read data from DRAMs arrived at ATE. As shown in the Figure 3, approximately 1ns gap occurs throughout DIMM.

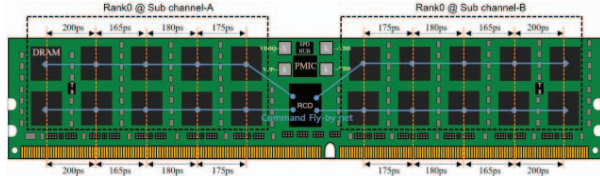


Figure 3. The operation delay due to fly-by topology of command and address nets in DDR5 RDIMM 2Rx4.

One of the most important elements of PI is di/dt , which is the amount of current change per unit time. Therefore, it can be expected to reduce and distribute the peak current that applying a gap of ~ 1 ns for each DRAM components respectively, which can result in preventing over estimation for the noise and the amount of the decoupling capacitor on DIMM. The Figure 4 shows the change in total current of DIMM when PCB delay between DRAMs is applied during write operation. Although the average level is almost same, peak level, which can be said to be an indicator of di/dt , is decreased by more than 20% from 5.6 to 4.5 Apk. So, we applied PCB delay to the PI simulation when summing each DRAM current profile on DIMM. As a result of comparing level of voltage noise based on the following results, it was confirmed that simulated voltage noise was almost same by less than 1% as shown in Figure 5, even though peak current level is reduced by 20%. In other words, a reduction in peak current by 20% has no change in noise and applying PCB delay to the PI simulation didn't make meaningful effects.

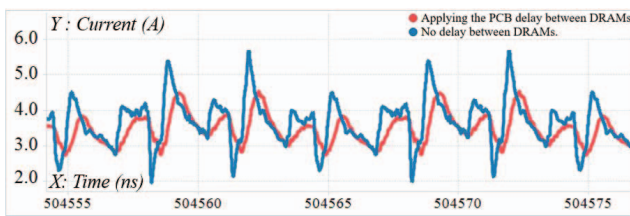


Figure 4. Effect of applying PCB delay between DRAMs for summing current profile.

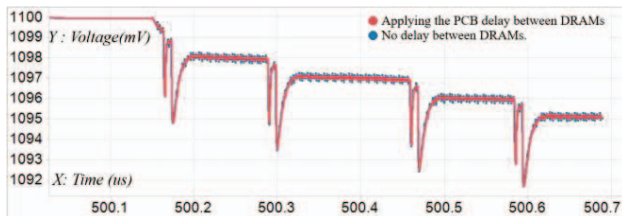


Figure 5. Comparison of voltage noises depending on applying PCB delay.

As result of analyzing the reason for that, it was verified as a phenomenon caused by the DDR5 PMIC operation. As mentioned in previous section, DDR5 DIMM adopts PMIC which supplies power to DRAM, RCD, TS and SPD on the DIMM. The DDR5 PMIC is the type of SMPS, charging and discharging repeatedly according to the switching frequency while supplying power throughout the DIMM as shown in Figure 6. What should be remembered is that the switching frequency range of DDR5 PMIC is from 500KHz to 1.25MHz. From PMIC point of view, the current change in ns scale as shown in Figure 4 is a microscopic world. In other words, PMIC cannot respond to such a fast current change like DRAM write or read operation, which results in voltage noise generated by DRAMs in a DIMM is mostly determined by slow and low frequency components of current changes. That is the reason why there is no difference depending on applying PCB delay to PI simulation.

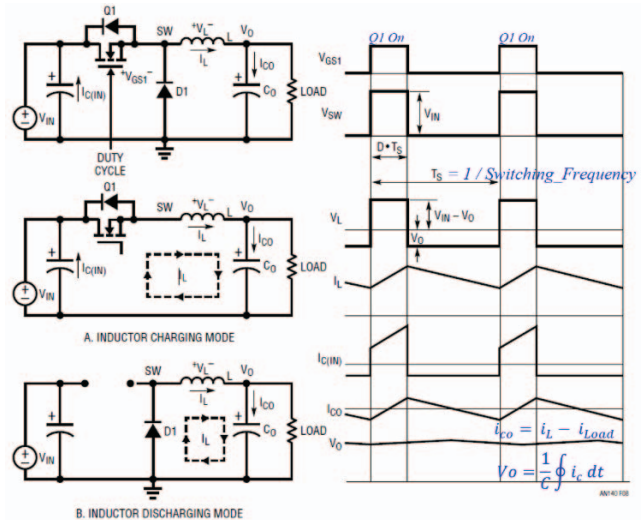


Figure 6. Buck converter type SMPS typical waveform [3].

B. Current Profile in CPM vs Average Current

To verify the analysis in the previous section, CPM-based simulation and average DC current-based simulation are compared with the same power scenario in this section. Figure 7 shows the comparison of 2 power models that has same average current. Red-line shown in Figure 7 is obtained by averaging current profile of Blue-line. Power scenario used in this simulation is implemented by the operation that DDR5 DRAM changes the operation from stand-by to bank-interleave write with 50A/us current slew. As a result of the comparison, simulated voltage noises are equal each other depending on 2 power model by 1~2mV as shown in Figure 8. When filtering noise under 20MHz is applied to the result in order to extract power noise components only generated by power changes, we could get almost same voltage noise waveform within 2%. Therefore, it can be seen that power noise by current changes in ns scale is almost filtered out since PMIC as a power source cannot react to that.

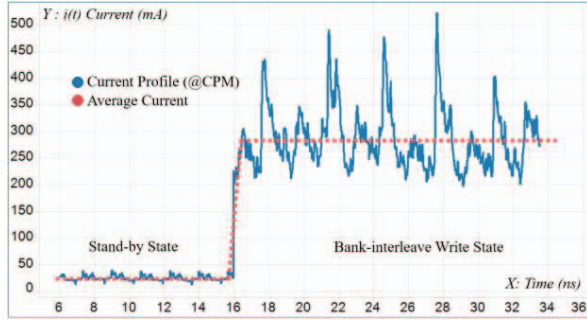


Figure 7. Comparison of type of power models (Current profile vs Average current)

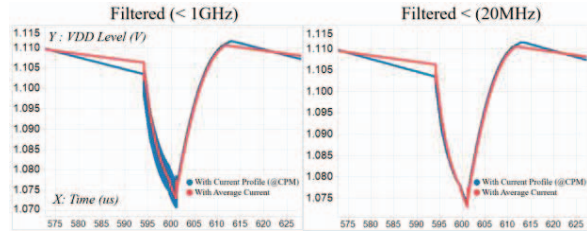


Figure 8. Comparison of voltage noise depending on power model (Current profile vs Average current)

C. Advantages of the Average current

Current profile and CPM, which are mainly used for PI simulation, are extracted by full circuit simulation which can take several weeks for just a few power scenarios. And it is very difficult to accurately reflect DRAM condition and setting to the simulation environment such as ODT (On-Die Termination), command gap, which makes getting consistency difficult. Moreover, the transient simulation time also takes more than 10~20 hours, making it impossible to simulate various power scenarios. Because the current profile is a PWL (Piece-Wise Linear) model in which the current is defined in every pico-second. On the other hand, average current can be measured and obtained even more easily than CPM. And average current depending on various operation mode of DRAM is defined in JEDEC specification. It is called IDD as shown in Figure 9. DRAM IDD data are sometimes provided to customers because it is measured in the standard condition defined in JEDEC DDR5 specification. IDD have several advantages compared to CPM. First, IDD can be easily and rapidly obtained by testing DRAM with JEDEC standard power scenario which can be easily implemented. Second, IDD is even lighter than CPM because it is average current and doesn't have time resolution unlike CPM. Third, various power scenarios can be simulated with IDD, while CPM-based simulation can verify just a few worst scenarios due to simulation time. Therefore, if IDD can be used in PI simulation on behalf of CPM, it can make PI simulation even faster. Also it can be used in guaranteeing various power scenarios by utilizing the speed of IDD based simulation. In other words, various DRAM power scenarios can be reflected in PI simulation by utilizing the speed of IDD based simulation and it can be used and implemented in transient voltage noise simulation

as mention in previous section. This is the main starting point for implementing MCSRPM.

TABLE I. JEDEC DDR5 IDD SPECIFICATION [4]

IDD	Description
IDD0	Operating One Bank Active-Pre-charge Current
IDD0F	Operating Four Bank Active-Pre-charge Current
IDD2N	Pre-charge Standby Current
IDD2NT	Pre-charge Standby Non-Target Command Current
IDD2P	Pre-charge Power-Down Current
IDD3N	Active Standby Current
IDD3P	Active Power-Down Current
IDD4R	Operating Burst Read Current
IDD4W	Operating Burst Write Current
IDD5B	Burst Refresh Current (Normal Refresh Mode)
IDD5F	Burst Refresh Current (Fine Granularity Refresh Mode)
IDD5C	Burst Refresh Current (Same Bank Refresh Mode)
IDD6N	Self-Refresh Current: Normal Temperature Range
IDD7R	Operating Bank Interleave Read Current
IDD7W	Operating Bank Interleave Write Current

IV. MARKOV-CHAIN-BASED STOCHASTIC RANDOM POWER MODEL

In this chapter, main idea is described how to reflect realistic power scenarios to PI simulation using Markov-chain modeling and IDD.

A. Ideation of MCSRPM

Like other electronic devices, DRAM has various operation modes and transition logics between states which are defined in JEDEC specification. For example, if a certain row in a bank is active it must first be pre-charged (PRE) before another row can be activated [5]. Also, a target bank must be activated in order to write data to the target bank in DRAM. And target banks must be inactivated in order to refresh all banks (REFab) or refresh same bank (REFsb) in DRAM. In other words, DRAM write operation cannot be performed unless the bank is active and refresh cannot be performed while target banks are activated. As such, there are pre-conditions that must be satisfied to enter each operating state of DRAM. It is described as a state diagram in the JEDEC Specification as shown in Figure 9. That is, DRAM operates according to its state diagram regardless of whether it operates in system or in ATE. As mentioned in previous section, it is verified that there is very little difference between CPM and IDD-based simulation in term of predicting voltage noise. Also, IDs already has various power scenarios since it is the average current depending on operating modes of DRAM. In this paper, we propose the simulation methodology which can reflect numerous operating scenarios in PI simulation, in which we connect IDs depending on operating modes with stochastic chain based on DRAM state diagram using Markov-Chain modeling which has been used for modeling chain reactions

in many fields. Figure 10 shows another example of Markov-Chain modeling of molecular communication system [6]. For applying it to PI simulation, we developed MCSRPM generator coded by PERL script that generates the random power model based on the probability setting between IDD states. As a result, MCSRPM can reflect realistic power scenario to PI simulation through programming the weight of stochastic chain between IDD states since MCSRPM generator operates based on DDR5 DRAM state diagram basically.

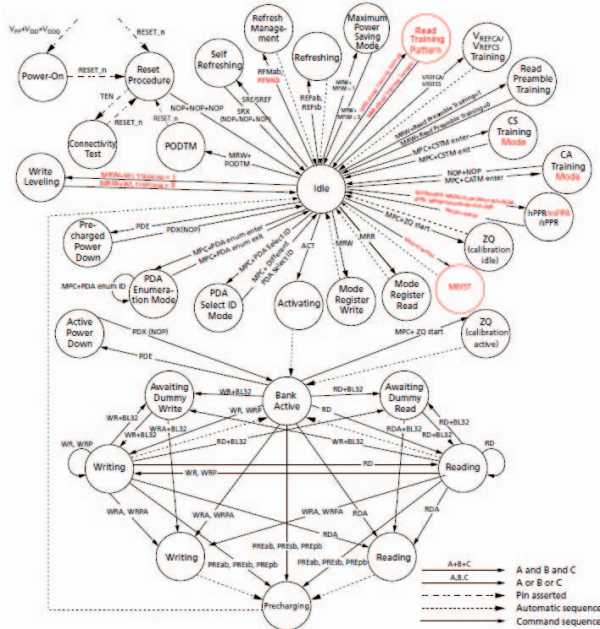


Figure 9. DDR5 DRAM state diagram [4].

B. Implementation of MCSRPM

As shown in Figure 11, weights are applied to chains between each IDD state based on DRAM state diagram. In order to reflect realistic power scenarios, it needs to be verified what kind of parameter should be randomly selected in MCSRPM generator. In this paper, we introduce 4 parameters randomly selected based on weight setting in MCSRPM generator.

1) *Scheduling IDD states*: As mentioned in previous section, IDD states are randomly scheduled by the weight that user set in MCSRPM generator. In other words, applying the weight to each IDD chain means the realistic DRAM power scenarios are reflected in PI simulation. For example, if user wants to apply more weight to write operation than read, user can assign a higher weight to IDD4W than IDD4R in MCSRPM generator.

2) *Current slew when changing IDD state*: Current slew when changing the IDD state was verified and extracted through circuit simulation and measurement. As a result, we implemented to select it randomly from 10 to 50A/us depending on weight user set.

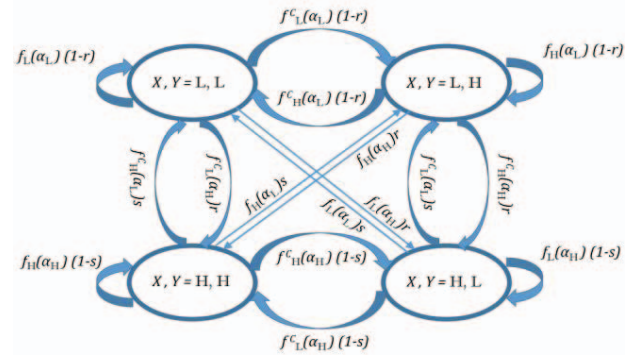


Figure 10. Examples of Markov-Chain modeling of molecular communication System [6].

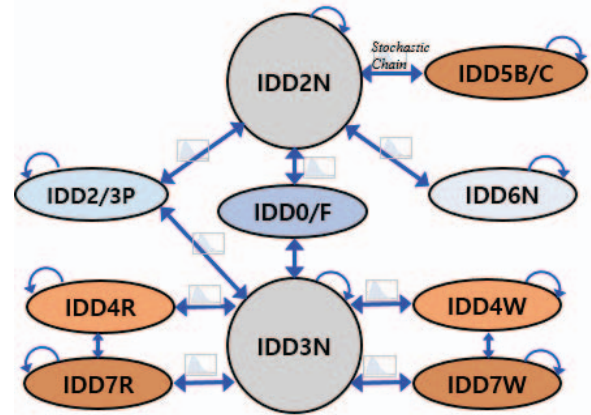


Figure 11. Stochastic IDD chains based on DRAM state diagram.

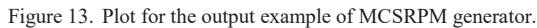
3) *Scaling factor modeling partial bank operation*: All IDD states defined in JEDEC specification make all banks operating in DRAM. However, memory operation in real system does not always make all banks operating in DRAM. Therefore, the scaling factor is randomly selected and multiplied to IDD states depending on the weight user set. Average number of the activated banks in system can be verified by measuring memory traffic with logic analyzer.

4) *Duration of IDD State*: The duration of the single IDD state is randomly selected according to the weight set by the user.

C. Result of MCSRPM

In this section, we introduce the output examples of MCSRPM generator. As mentioned in previous section, MCSRPM generator works with IDD and the weight set by user. And then, it generates the stochastic power model with randomly selected 4 parameters. Figure 12 shows output example which is in SPECTRE sub-circuit format. There is no time resolution unlike current profile because MCSRPM consists of multiple IDD states, which makes simulation even faster. Regarding the speed, it takes 5 seconds to generate the random power model which has 80,000 IDD states. Also simulation time with MCSRPM which has 5,000 power states takes 316 minutes while CPM based simulation with only 2 power scenarios takes 780 minutes.

Figure 12. Output example of MCSRPM generator.



In this chapter, we verify the effectiveness of the proposed MCSRPM and how to implement system power scenario in MCSRPM. As mentioned in previous section, MCSRPM is a type of power model generated by changing the power state with the weight between IDD states based on DDR5 state diagram. Therefore, system singularity indexes were extracted as a form that is easy to implement in the weight between IDD states in order to effectively reflect system power scenario to MCSRPM. As a result of that, we verified and predicted with more than 90% consistency in terms of the voltage peak to peak noise measured in DDR5 system.

At first, memory traffic was measured with a logic analyzer while operating various test programs in the DDR5 system and the software called “System Memory Traffic Analyzer (SMTA)” was developed to extract and calculate the singularity of memory traffic in the system. In this paper we introduce 3 indexes that can best reflect the system power scenarios in MCSRP.

be calculated by the number of write and read commands in the memory traffic data measured with logic analyzer.

2) *The number of activated banks*: It indicates how many banks are activated among total 32 banks in DRAM over the entire time. It is used to calculate the weight of partial bank scaling factor mentioned in previous section.

3) *Markov-Chain command table*: This table represents scheduling DRAM commands by CPU(Central Processing Unit) as stochastic chain. Overall scheduling could be predicted and the portion of operating modes could be also predicted such as the portion of write and read operation.

In this section, we verify the consistency between transient simulation results of MCSRPM and measurement in DDR5 system in terms of voltage peak to peak noise. First, 3 singularity indexes were extracted from measured memory traffic depending on test programs in DDR5 system. And then we analyze singularities with SMTA. In the case of Program-B, which is a kind of memory test program, data bandwidth is confirmed to be about 70.2% as shown in Figure 14 and the average number of activated banks is about 11 as shown in Figure 15. Table II shows Markov-Chain command table which expresses overall command scheduling as a form of stochastic chain. For example, the probability of going from “Active” to “Read” is 37% in Table II.

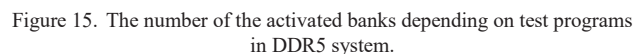
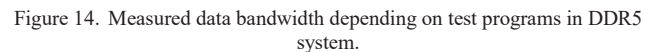


TABLE II. MARKOV-CHAIN COMMAND TABLE DEPENDING ON TEST PROGRAM IN DDR5 SYSTEM

Test PGM (Option)	Prev. CMD \ Curr. CMD	ACT	WR	RD	PRE	PREsb	REFsb
Program-A	Active	12%	32%	22%	33%	1%	0%
	Write	50%	14%	0%	34%	1%	1%
	Read	27%	0%	39%	33%	1%	1%
	Pre-charge	38%	22%	21%	18%	0%	1%
	PREsb	33%	16%	18%	33%	0%	1%
	REFsb	0%	33%	42%	25%	0%	0%
Program-B	Active	10%	21%	37%	31%	1%	0%
	Write	49%	15%	0%	35%	1%	1%
	Read	35%	0%	26%	38%	1%	1%
	Pre-charge	38%	15%	27%	19%	0%	0%
	PREsb	38%	12%	21%	28%	0%	0%
	REFsb	0%	26%	53%	21%	0%	0%
Program-C	Active	10%	16%	44%	29%	1%	0%
	Write	48%	17%	0%	33%	1%	1%
	Read	36%	0%	23%	40%	1%	1%
	Pre-charge	33%	10%	34%	22%	0%	0%
	PREsb	42%	9%	25%	24%	0%	0%
	REFsb	0%	19%	61%	20%	0%	0%
Program-D	Active	4%	0%	78%	17%	1%	0%
	Read	35%	0%	40%	23%	0%	1%
	Pre-charge	25%	0%	59%	15%	0%	0%
	PREsb	41%	0%	45%	14%	0%	0%
	REFsb	0%	0%	94%	6%	0%	0%

TABLE III. MARKOV-CHAIN IDD TABLE

Previous Power State \ Current Power State	IDD0	IDD0F	IDD2N	IDD2NT	IDD2P	IDD3N	IDD3NT	IDD4R	IDD4W	IDD5B	IDD5C	IDD5F	IDD6N	IDD7R	IDD7W
IDD0	2	0	0	0	0	9	0	0	0	0	0	0	0	0	0
IDD0F	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
IDD2N	9	0	3	0	0	0	0	0	0	0	1	0	0	0	0
IDD2NT	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
IDD2P	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0
IDD3N	0	0	0	0	0	0	0	6	3	0	0	0	0	4	2
IDD3NT	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
IDD4R	0	0	2	0	0	3	0	9	4	0	0	0	0	4	2
IDD4W	0	0	2	0	0	3	0	6	3	0	0	0	0	4	2
IDD5B	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
IDD5C	0	0	9	0	0	0	0	0	0	1	0	0	0	0	0
IDD5F	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
IDD6N	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0
IDD7R	0	0	2	0	0	0	0	6	3	0	0	0	0	4	2
IDD7W	0	0	2	0	0	0	0	6	3	0	0	0	0	4	2

In the case of Program-B, the ratio of read to write can be calculated and it is 44.3:25.9 which is equivalent to 1.7:1. To implement the ratio of read to write and data bandwidth, the weight of read ratio needs to be set higher than that of write, making data bandwidth about 70% as shown in Table III. For example, the probability of going from IDD4R to IDD4W can be calculated by $4/(2+3+9+4+4+2)$ as shown in Table III. Also, the probability is applied only to IDD5C,

not to IDD5B and IDD5F, since we observed that DDR5 system we used employed only same bank refresh mode through analyzing Table II. Second, we extract singularities of memory scheduling in system from measured memory traffic data and apply them to random power model which is created with stochastic IDD chain table, randomly selected current slew and duration in MCSRPM generator. After that, it need to be verified how well created MCSRPM implements system singularities. Figure 16 shows the portion of IDD states in the created MCSRPM. As a result, the ratio of read to write and data bandwidth in MCSRPM are properly implemented as we intended.

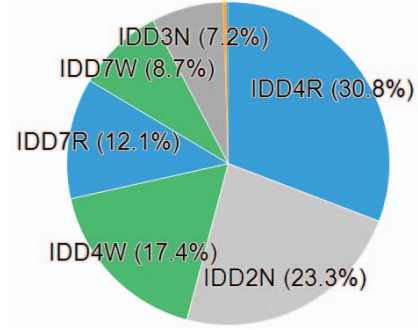


Figure 16. The number of the activated banks depending on test programs in DDR5 system.

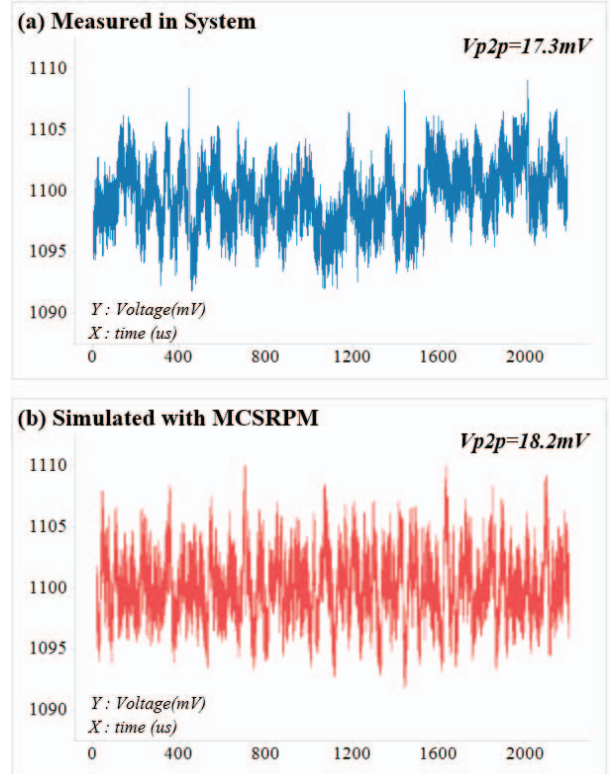


Figure 17. Comparison between measured voltage noise (a) and transient simulation result with MCSRPM (b).

Finally, we simulate the voltage noise with implemented MCSRPM which reflects system singularities of Program-B modeled by probability and Markov-Chain. In this simulation, we use SPECTRE simulation tool in order to generate the voltage noise and then compare it with measured voltage noise in a system environment that 2Rx4 RDIMMs are being operated by Program-B in DDR5 system. As a result of comparing them, we confirmed that the output of MCSRPM and the measured noise are correlated at more than 90% in terms of voltage peak to peak noise as shown in Figure 17. Moreover, 90~95% consistency is verified continuously even if the same experiment is repeated. The shape of noise are a little bit different each other. If the IDD scenarios are further subdivided than those defined by JEDEC, both the amount of voltage noise and shape can be predicted more accurately. Because, it is easy to implement subdivided IDD scenarios with ATE or other testers.

CONCLUSION

In this paper, the PI simulation method was proposed for reflecting system scenario and reducing simulation time. The proposed method had numerous advantages in terms of implementation and effectiveness. First of all, we verified average current was effective in PI simulation due to the low switching frequency of PMIC. Second, the effectiveness was verified for using IDD's which is a kind of average current depending on operation and power states in DRAM. Third, we connected IDD's with the chain, in which we applied probability to the chain between IDD's in order to reflect power scenario in an actual DDR5 system. Fourth, we developed the software which could analyze singularities of system scheduling from the measured memory traffic and they were applied to MCSRPM generator. Finally, the voltage noise simulated with implemented MCSRPM is correlated with the measured noise at more than 90%. This paper proved that tons of system power scenarios could be reflected in existing environment of PI simulation using stochastic and Markov-Chain modeling. It is also verified its practicality through making simulation time reduce by 60% compared to CPM-based simulation which could verify just a few power scenarios.

REFERENCES

- [1] Q.Chen. et al., Design and Demonstration of Low cost, Panel-based Polycrystalline Silicon Interposer with Through-Package-Vias(TPVs). IEEE Electro. Components Technol. Conf. (2011)
- [2] J. J. Kim et al., "A Novel Dynamic PI Simulation Methodology Including Non-linear PMIC Characteristics for DDR5 Module," 2019 Electrical Design of Advanced Packaging and Systems (EDAPS), 2019, pp. 1-3, doi: 10.1109/EDAPS47854.2019.9011619.
- [3] Henry J. Zhang, Basic Concepts of Linear Regulator and Switching. Linear Technology Application Note (2013)
- [4] JEDEC DDR5 Full Specification Draft Rev1.71
- [5] L. Steiner, M. Jung and N. Wehn, "Exploration of DDR5 with the Open-Source Simulator DRAMSys," MBMV 2021; 24th Workshop, 2021, pp. 1-11.
- [6] M. Ahuja, Ankit and M. R. Bhatnagar, "Markov Chain Modeling of the End-to-End Molecular Communication System using Ligand